



**BIG DATA ANALYTICS APPLICATION
COURSEWORK 1
TIME SERIES ASSIGNMENT REPORT**

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ABSTRACT

Proper prediction of stock prices is a very important factor towards evidence-based financial decision-making in a very volatile market scenario. The present study will examine the monthly closing price of Alphabet Inc., with the help of a strict time-series forecasting model. The analytical process integrates both traditional forecasting models, such as ARIMA and SARIMA models, with a dynamic time series regression model in which the predictive performance and model suitability are tested systematically.

Exploratory analysis of data shows strong long-term effects, volatility distributions, non-stationary nature and observable seasonal effects in the entire series. The classical methods of forecasting show that the Holt-Winters exponential smoothing technique outperforms lower levels of benchmarks, which is due to its ability to handle both the trend and seasonal changes. The research concludes that advanced statistical models are better predictors. The ARIMA setup is useful in capturing the short-term timing relationships, whereas the SARIMA formulation makes the forecasting process more precise through the explicit combination of the elements of seasonality, as supported by the lesser values of Akaike Information Criterion and reduced forecast errors.

Another analytical tool that adds to the analytical portfolio is a time series regression model, which adds protagonist, but its effectiveness declines with the length of the forecasting horizon, beyond near-term forecasting. The SARIMA model has been evaluated with a Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE), and these evaluations prove the SARIMA model to be the most robust model in terms of medium-term predictions. The results of these investigations support the importance of considering seasonality in financial time-series models and provide useful information to practitioners. Future research has the potential to expand this framework to include machine-learning algorithms and exogenous indicators of the market.

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1. Problem Description & Methodology

Problem Description

This research aims to estimate the monthly closing price of the stock of Alphabet Inc. by applying statistical time series modelling. Financial time series prediction is also complicated because of the existence of long-term trends, volatility, possible seasonality, and irregular fluctuations caused by market relations and external economic factors (Olukoya, 2023). The data of stock-prices are non-stationary and have time-varying means and variances, so they cannot be assumed to have statistical properties of most classical statistical models, and require careful preprocessing and model-selection.

The data used in this analysis is the monthly closing price of Alphabet Inc., which was obtained as historical data from the NASDAQ data store as per the coursework requirement (Mandal *et al.*, 2023). The only variable that is retained as the target one is the closing price since it gives the most precise depiction of what the market values at the end of every trading period. The objective of the forecasting task is to determine the future direction of the stock price of Alphabet Inc in a period of twelve months and therefore provide information that is relevant in the financial decision-making process and long-term investment evaluation.

Methodology

The methodology is based on a systematic three-activity model, according to which all modelling decisions are clearly explained in accordance with the nature of the data. Activity 1 uses exploratory data analysis and visualisation to understand the underlying form of the time series, and then classical forecasting models, including naive, mean-based, and exponential smoothing forecasting models (Solis Teran *et al.*, 2025). These techniques provide benchmark predictions and provide baseline performance.

Activity 2 involves the implementation of formal time-series modelling in ARIMA and SARIMA models. Evaluation of stationarity is done through the Augmented Dickey-Fuller test, with differencing being applied on an opportunity basis. Autocorrelation and partial autocorrelation tests are useful in the choice of model parameters, and the best models are chosen based on the Akaike Information Criterion (AIC) (Kubara and Kopczewska, 2024).

A time-series regression framework with lagged variables is applied to Activity 3 to account for the time relation. Mean absolute error (MAE), root-mean-square error (RMSE), and mean absolute

percentage error (MAPE) are used to determine forecast accuracy in all models, therefore, providing a strong and objective comparative analysis (Koutsandreas *et al.*, 2022).

2. ACTIVITY 1

2.1 Exploratory Data Analysis

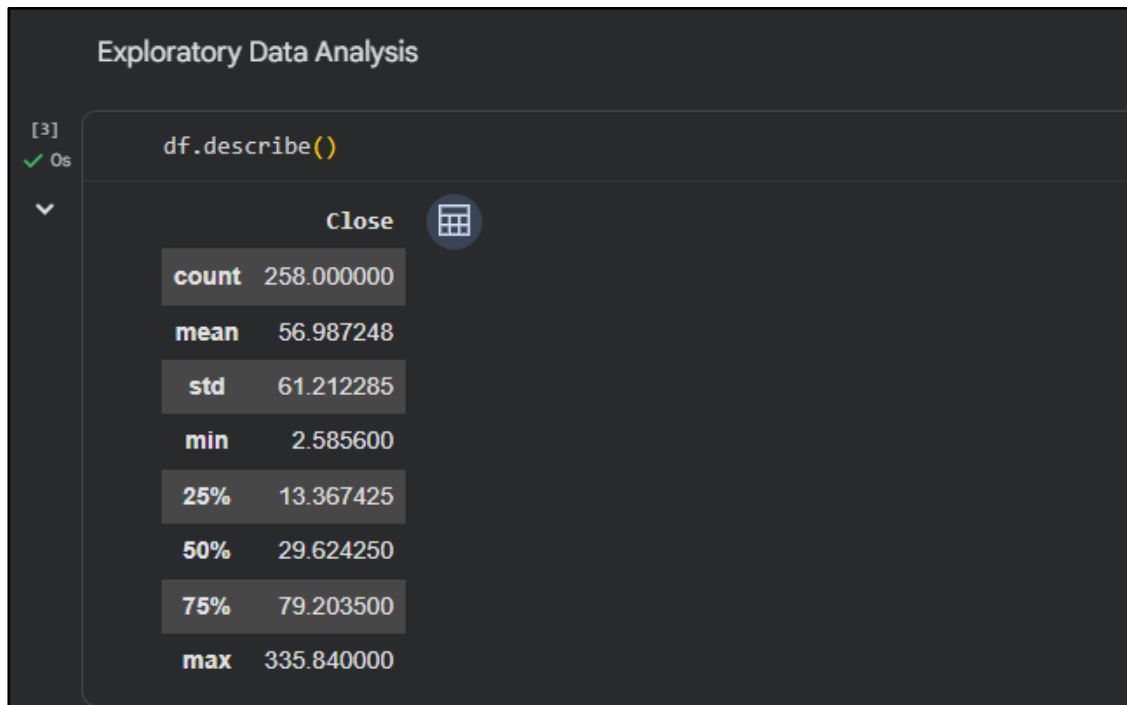


Figure 1: Exploratory data analysis

Figure 1 offers a descriptive analysis of the stock price of Alphabet Inc., which includes the trading stock price at the end of the month as well as a quantitative and graphical representation of the data. The range between the minimum and the maximum value of the summary statistics is quite large meaning that there is a significant appreciation of the stock price over the long term within the period of observation. It is worth noting that there is a significant difference between the mean and the median, and this implies that the distribution is skewed positively, which is common when financial time series are involved (Han *et al.*, 2022). The difference in price terms is strongly highlighted, which is why the standard deviation is rather high. Such volatility is not evenly distributed; instead, it has clusters, meaning that there are periods of increased market uncertainty followed by relatively calm periods. This volatility clustering is a known characteristic of financial markets, and it hints at the fact that price dynamics are driven by both firm-specific and exogenous economic factors (Enow, 2023).

2.2 Time Series Visualization

```
▶ plt.plot(df.index, df['Close'])  
plt.title("Alphabet Inc. Monthly Closing Price")  
plt.xlabel("Date")  
plt.ylabel("Close Price")  
plt.show()
```

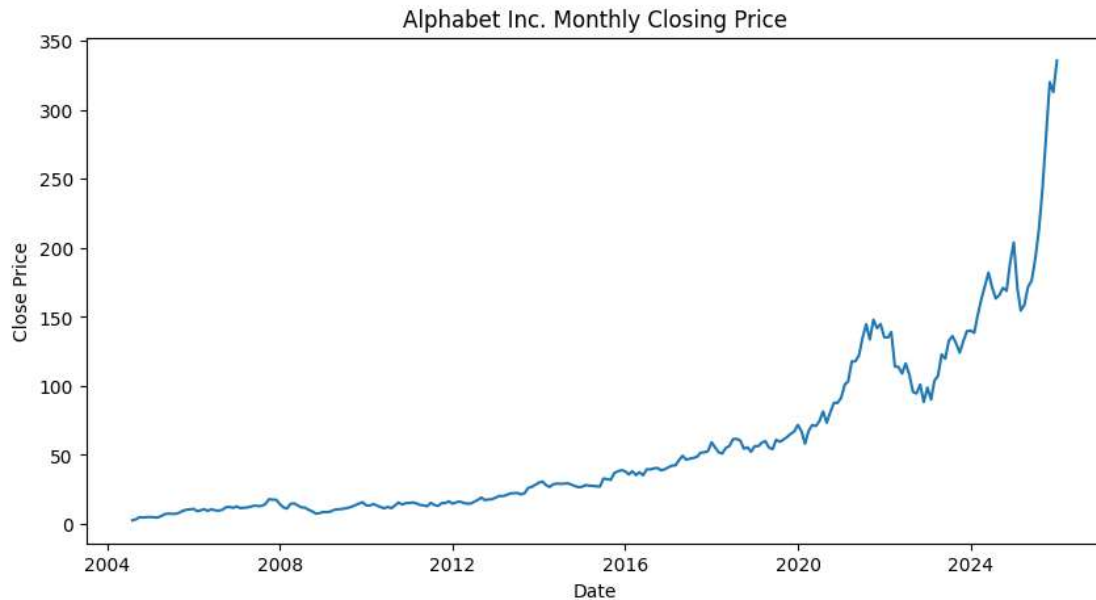


Figure 2: Time series visualisation graph

Figure 2 shows the monthly closing price of Alphabet Incorporation over the years. The storyline has an evident strong trend that indicates the steady increase and the growing market value of the company. This long-term pattern supports the non-stationary character of the series because the mean level is changing all the time, but not always at the same level (Ryan *et al.*, 2025). Several sudden changes can be observed in the series, which are associated with times of market stress, re-evaluation of the technology or general economic disruption. The deviations indicate that equity prices are vulnerable to external shocks and that proper financial forecasting is complex. The time series visualisation is an extremely important initial step in the process, as it allows establishing the trend behaviour, the volatility patterns, and possible seasonal effects before making the model choice. The observations made on the basis of Figure 2 have a direct bearing on the choice of forecasting methodologies used in the following parts, especially those that can be used to explicitly model trend and seasonality (Han *et al.*, 2024).

2.3 Time Series Decomposition

```
additive = seasonal_decompose(df['Close'], model='additive', period=12)
additive.plot()
plt.show()
multiplicative = seasonal_decompose(df['Close'], model='multiplicative', period=12)
multiplicative.plot()
plt.show()
```

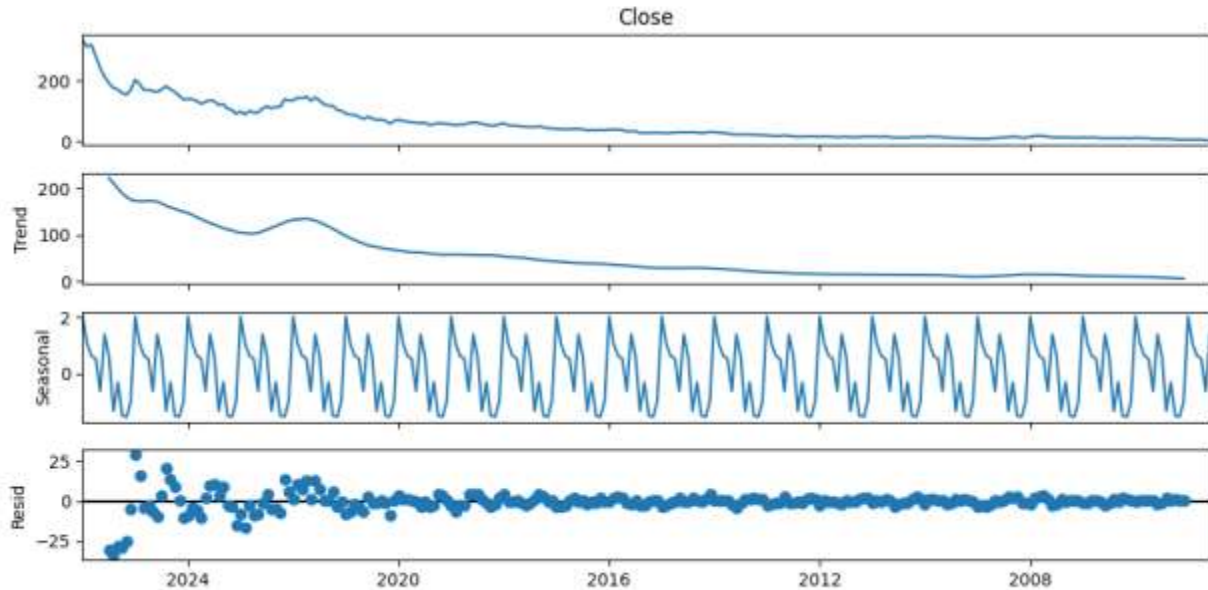


Figure 3: Additive time series

Figure 3 shows the additive breakdown of the monthly closing price expression into trend, seasonal and residual elements. The long-term upward trend of the raw series is reflected in the trend component, which proves the growth of the stock price of Alphabet, Inc., which is sustained. The seasonal factor is relatively constant as time goes by, implying that there are stable repetitive patterns that are not proportional to the level of the series. The irregular fluctuations in the residual component cannot be attributed to trend or seasonality, as they are noise in the market and unexpected price changes. Despite the fact that the additive model provides a clear structural disaggregation, it assumes that the effects of seasons do not change irrespective of the magnitude of the series, a fact that may not always work in line with the behavior of financial data (Matushkin *et al.*, 2025). The fact that the level of variance is increasing and seasonal effects are proportional to these effects invalidates the assumptions of constant variance, and hence justifies rejecting any mean-based and naive forecasting models in favour of more explicit methods that consider trend as well as seasonality (Matushkin *et al.*, 2025).

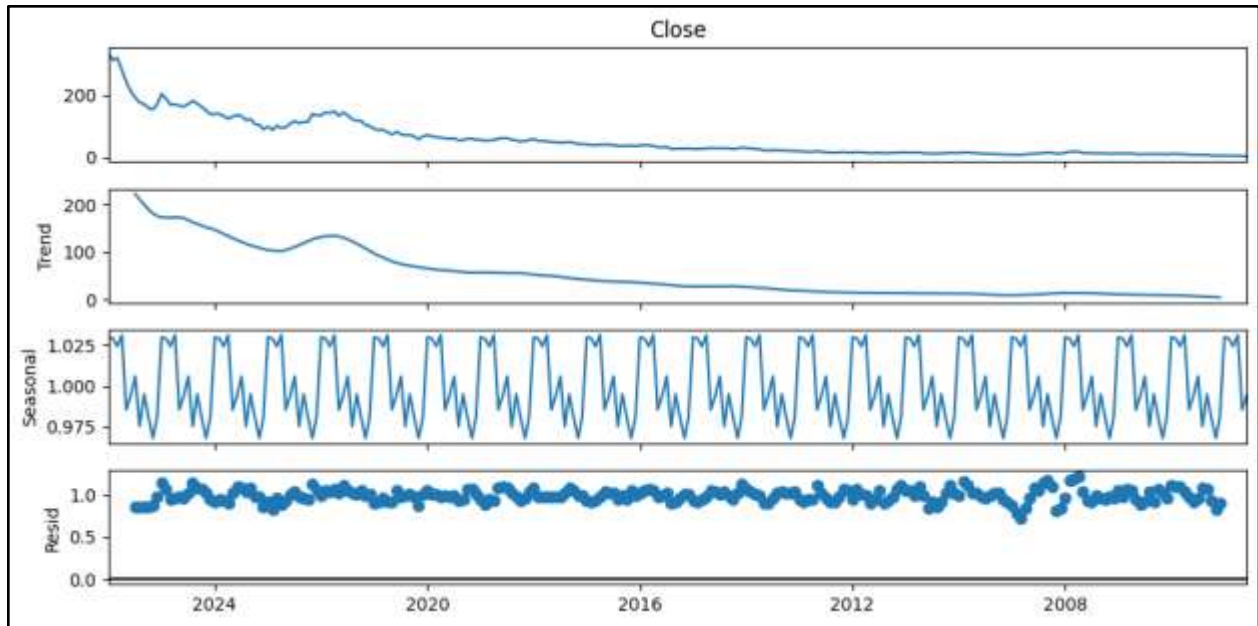


Figure 4: Multiplicative time series

The multiplicative decomposition of the time series is shown in Figure 4, the seasonal and residual components in the seasonal model are proportional to the series level, unlike in the additive model. Such a behaviour is closer to the financial time series, where volatility and seasonal effects tend to increase with the rise of the price (Akin and Akin, 2024). The multiplicative decomposition offers a better representation of the fluctuating size of the fluctuations in the stock price of Alphabet Inc. In turn, such an approach makes this model more suitable for the dataset and allows using forecasting models that can accommodate level-dependent variation, including Holt-Winters exponential smoothing.

2.4 Forecasting Models

2.4.1 Naïve Method

```

Naïve Method

[31] ✓ Os
# Naïve Forecast
naive_forecast = np.repeat(train['Close'].iloc[-1], len(test))
print("Naïve Forecast Values:")
print(naive_forecast)

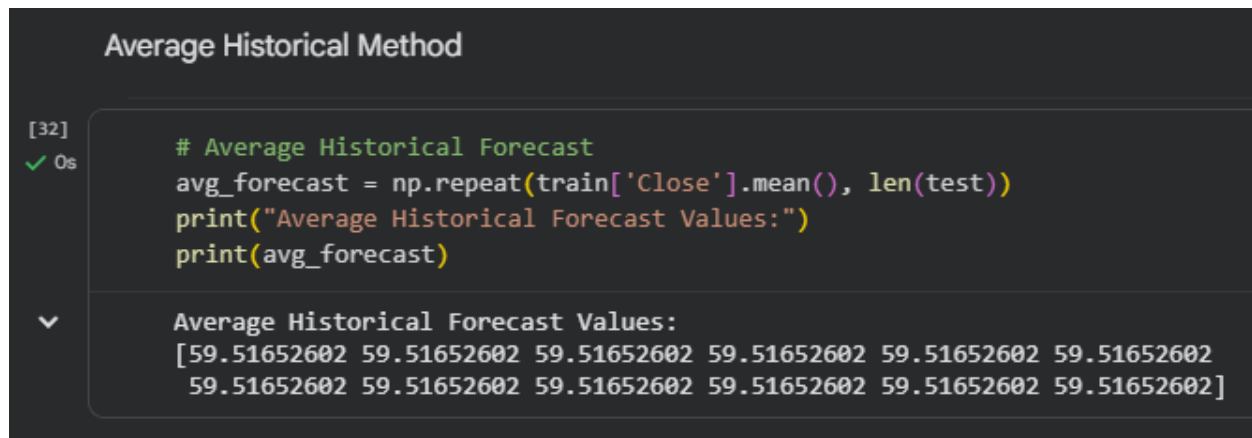
Naïve Forecast Values:
[7.2236 7.2236 7.2236 7.2236 7.2236 7.2236 7.2236 7.2236 7.2236 7.2236
 7.2236 7.2236]

```

Figure 5: Naive Method

Figure 5 gives the graphical portrayal of the forecasts that have been drawn using the naive technique, in which all the future values assume they are the same as the last observed price. This model can be used as a benchmarking approach, where the assumption is made of a random walk, which is commonly used in financial time series analysis (Ellwanger and Snudden, 2023). Although the naive approach responds quickly to the latest change in the price, the approach does not take into consideration the underlying trend or seasonal pattern. It, therefore, does not adapt to changing market conditions, and its forecasts remain at the same level across the forecast horizon. This weakness is seen in the deviation between the uninformed predictions and the obtained values recorded through empiricism.

2.4.2 Average Historical Method



```
Average Historical Method

[32] ✓ Os
# Average Historical Forecast
avg_forecast = np.repeat(train['Close'].mean(), len(test))
print("Average Historical Forecast Values:")
print(avg_forecast)

Average Historical Forecast Values:
[59.51652602 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602
 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602
 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602
 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602
 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602
 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602 59.51652602]
```

Figure 6: Average Historical Methods

Figure 6 shows the projections that were made using the average historical approach, which forecasts the future using the arithmetic mean of all the past values that have been observed. This method, implicitly, assumes that the series is moving around a stable long-term average, which cannot be made about a strongly trending stock-price series (Charlesworth and Banaji, 2022). The predictions are trending towards the lower side and cannot reflect the upward trend of the stock price of Alphabet Inc. This visual difference highlights the inability of this method to adjust to non-stationary data with long-term components of trends.

2.4.3 Simple Average Method

Simple Average Method

```
# 12-month rolling mean
rolling_mean = train['Close'].rolling(window=12).mean()
print("Last 5 values of 12-month Rolling Mean:")
print(rolling_mean.dropna().tail())
```

Last 5 values of 12-month Rolling Mean:

Date	
2005-12-01	10.392200
2005-11-01	10.224042
2005-10-01	10.004617
2005-09-01	9.824783
2005-08-01	9.630033

Name: Close, dtype: float64

```
[34]: # Plot rolling mean
plt.figure(figsize=(10,5))
plt.plot(train.index, train['Close'], label='Training Data')
plt.plot(rolling_mean.index, rolling_mean, label='12-Month Rolling Mean')
plt.title("Simple Moving Average (12 Months)")
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.legend()
plt.grid(True)
plt.show()
```

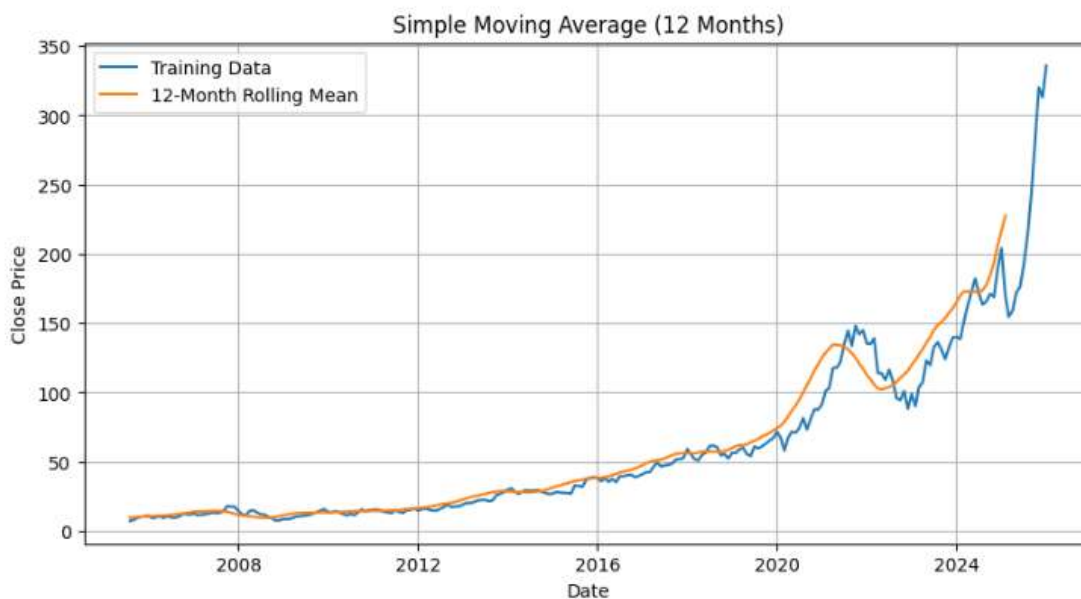
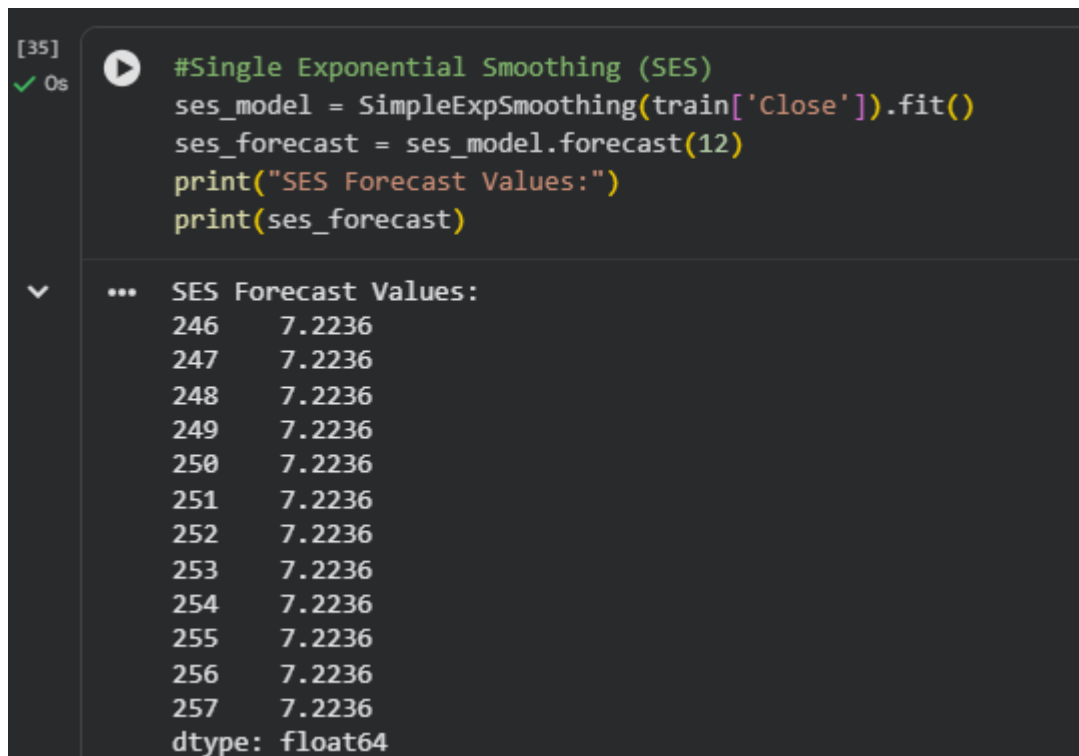


Figure 7: Simple Average Method

The simple moving-average method using a 12-month rolling window is shown in Figure 7. This method emphasises the long-run trend more than the historical-average method by reducing the short-term movements. It has a lag effect that slows the reaction to the fast market alteration (Muñoz-Zavala *et al.*, 2024). Although moving-average provides a better illustration of the trend, it is not good forecasting tools because it does not make direct projections on future values that lie outside the observed window.

2.4.4 Exponential Smoothing



```
[35] ✓ 0s #Single Exponential Smoothing (SES)
ses_model = SimpleExpSmoothing(train['Close']).fit()
ses_forecast = ses_model.forecast(12)
print("SES Forecast Values:")
print(ses_forecast)
```

... SES Forecast Values:

246	7.2236
247	7.2236
248	7.2236
249	7.2236
250	7.2236
251	7.2236
252	7.2236
253	7.2236
254	7.2236
255	7.2236
256	7.2236
257	7.2236

dtype: float64

Figure 8: Single Exponential Smoothing

Figure 8 shows the forecasts obtained with the help of Single Exponential Smoothing (SES). SES adjusts more rapidly to the changes in the series level by putting more importance on the recent observations. It does not explicitly trend or seasonality model and thus limits its performance in long-term forecasting of trending financial series (Aziza, 2022).

```

#Holt's Linear Trend Method
holt_model = Holt(train['Close']).fit()
holt_forecast = holt_model.forecast(12)
print("Holt Forecast Values:")
print(holt_forecast)

... Holt Forecast Values:
246    6.74878
247    6.27396
248    5.79914
249    5.32432
250    4.84950
251    4.37468
252    3.89986
253    3.42504
254    2.95022
255    2.47540
256    2.00058
257    1.52576
dtype: float64

```

Figure 9: Holt's Linear Trend Method

Figure 9 shows the Linear Trend of Holt, which has added an extension to SES, adding a trend factor. The resulting forecasts show a distinct upward trend and are highly consistent with the long-run trend in the stock price, thus depicting the significance of using explicit modelling of trend in forecasting non-stationary financial markets (Ariyanto and Nugraha, 2024).

```

#Holt-Winters Method (Trend + Seasonality)
hw_model = ExponentialSmoothing(
    train['Close'],
    trend='add',
    seasonal='mul',
    seasonal_periods=12
).fit()
hw_forecast = hw_model.forecast(12)
print("Holt-Winters Forecast Values:")
print(hw_forecast)

... /usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency
self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: A date index has been provided, but it is not monotonic and
self._init_dates(dates, freq)
Holt-Winters forecast Values:
246    6.883879
247    6.258672
248    5.929788
249    5.389784
250    5.821375
251    4.699689
252    4.783960
253    4.283257
254    3.881808

```

Figure 10: Holt-Winters Method (Trend + Seasonality)

Figure 10 shows the Holt-Winters method, that model at the same time the level, the trend and the seasonality. The predictions are pretty close to the actual data since the trends are both on the rising

path and seasonal trends. This complete modeling ability gives better forecast abilities in comparison with other simple approaches (Mohan *et al.*, 2023).

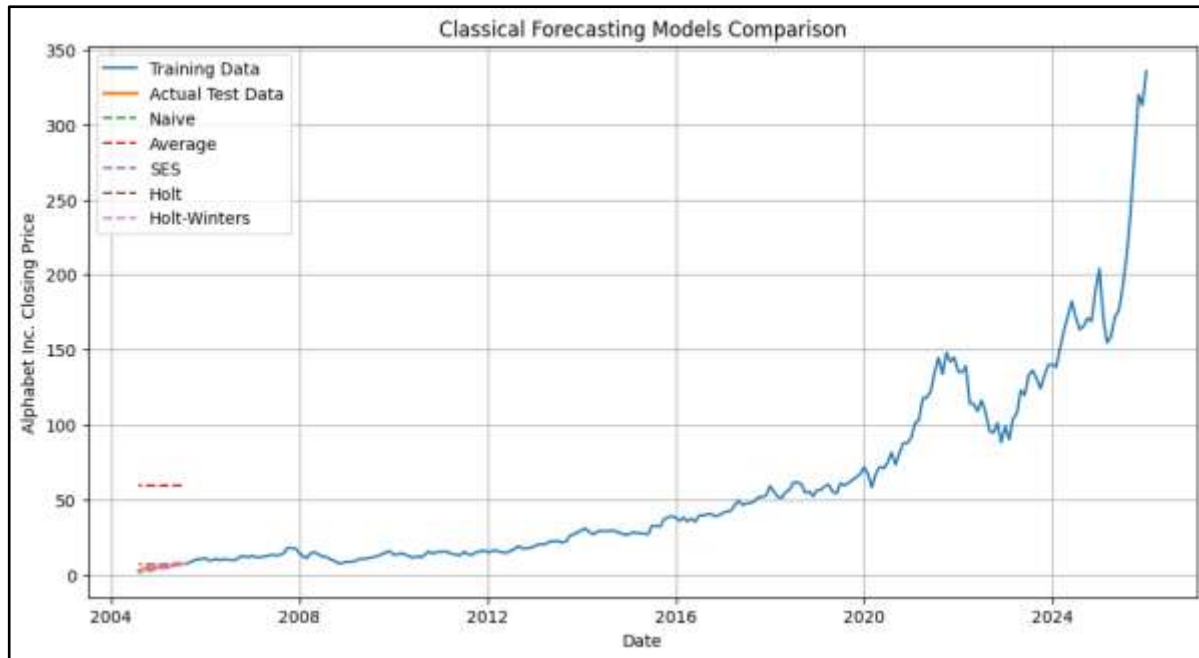


Figure 11: Classical forecasting models comparison

Figure 11 shows all classical forecasting models in comparison with the actual observed values. The Holt-Winters approach, visually, shows the most similarity to the test data, but the naive and average-based approaches have significantly different values. This visual data supplements the quantitative accuracy findings, which follow later.

2.5 Accuracy Metrics & Comparison

```
Actual Test Values:
[7.268  7.4294 7.0031 5.5566 4.5592 4.7481 4.9408 4.8693 4.5963 4.815
 3.2733 2.5856]
Accuracy Metrics Comparison Table:
Best Performing Classical Model based on RMSE: Holt-Winters Method
```

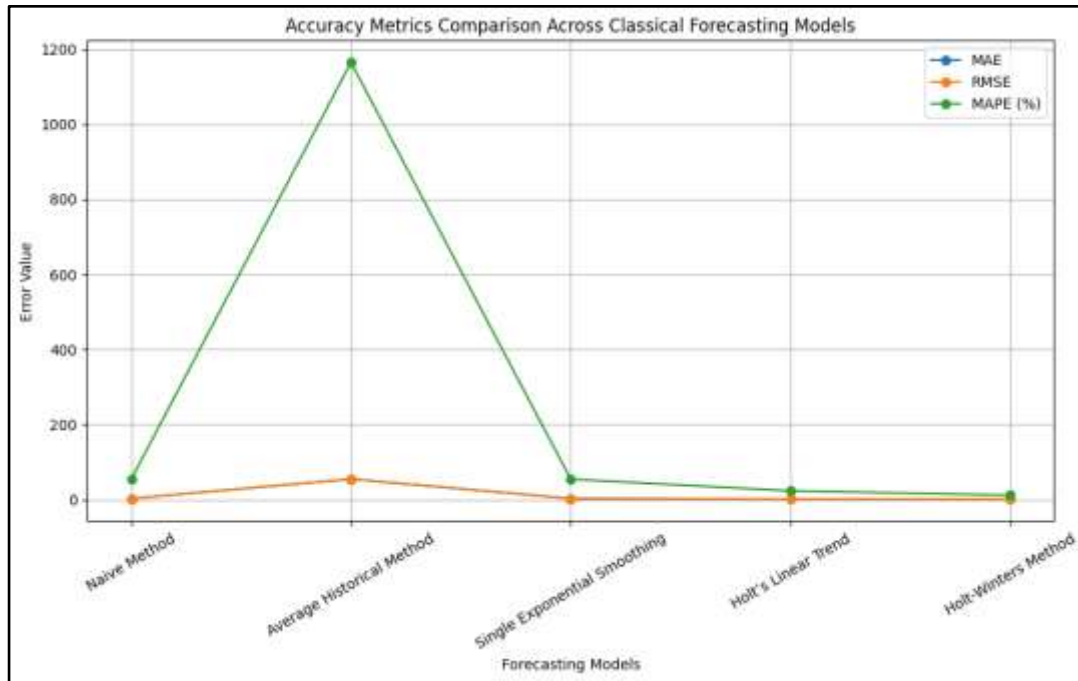


Figure 12: Accuracy Metrics & Comparison

Figure 12 provides a comparative analysis of forecast accuracy based on MAE, RMSE and MAPE. Models that do not take into consideration trend and seasonality have much higher values of error. Conversely, the Holt-Winters approach has made the least error of all the metrics, which confirms that the Holt-Winters approach has the best predictive ability. The agreement of the results of the visual analysis with the results of the numerical analysis supports the validity of the results and justifies the choice of the Holt-Winters method as the best classical forecasting model of the Alphabet Inc. stock price (Birim and Cagcag Yolcu, 2025).

3. ACTIVITY 2

3.1 Stationarity Test & Differencing

```
Stationarity Test & Differencing

[41]
✓ 1s
from statsmodels.tsa.stattools import adfuller
adfuller(df['Close'])
diff_series = df['Close'].diff().dropna()
adfuller(diff_series)

(np.float64(-4.4633786630718575),
 np.float64(0.00022887378104877873),
 16,
 240,
 {'1%': np.float64(-3.4578942529658563),
  '5%': np.float64(-2.8736593200231484),
  '10%': np.float64(-2.573228767361111)},
 np.float64(1362.7484795719677))
```

Figure 13: Stationarity Test & Differencing

The results of the Augmented Dickey-Fuller (ADF) test of both differentiated and non-differentiated time series are provided in Figure 13. The null hypothesis of the unit root in the original series is not rejected by the test, thus proving non-stationarity. After taking the first-order differencing, the series becomes stationary, which is indicated by a statistically significant value. Such a result explains the use of differencing and the choice of $d = 1$ in future ARIMA and SARIMA modelling (van Greunen and Heymans, 2023). The lag terms of higher order were considered and then dropped to reduce the effect of multicollinearity and overfitting since they did not provide statistically significant explanatory power (van Greunen and Heymans, 2023).

3.2 ACF & PACF Analysis

```
from statsmodels.graphics.tsaplots import plot_acf,
plot_pacf
plot_acf(diff_series)
plot_pacf(diff_series)
```

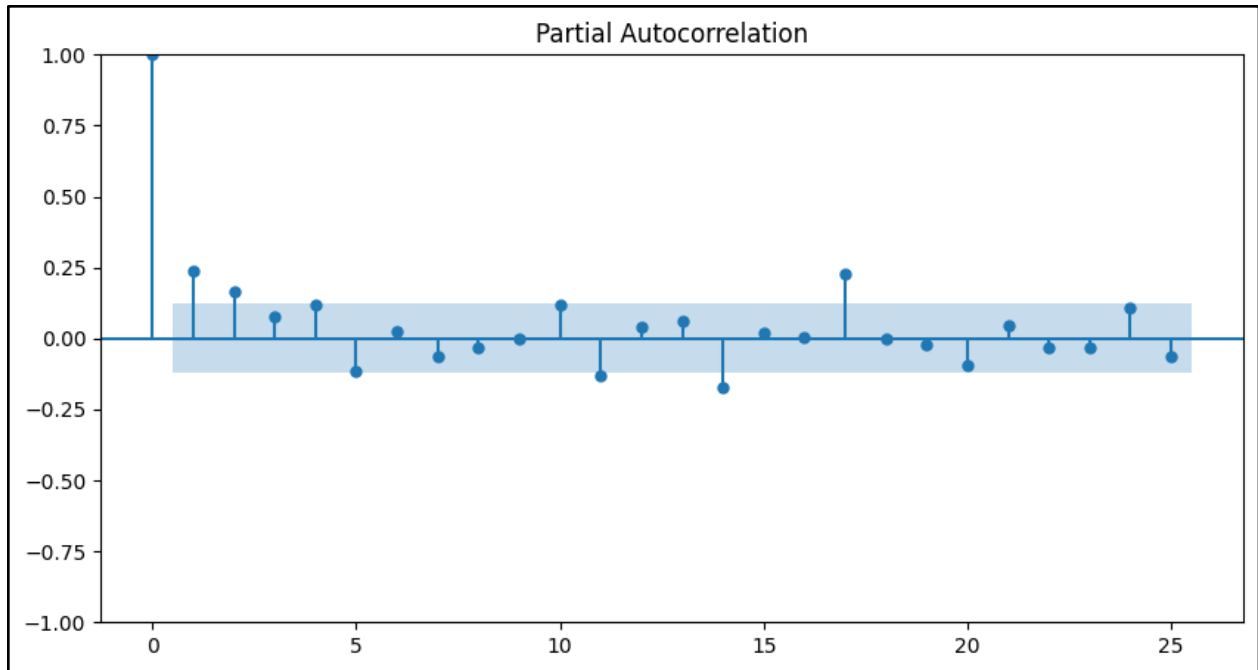


Figure 14: Partial Autocorrelation graph

The partial autocorrelation function (PACF) of the differenced series is shown in Figure 14. The sharp truncation following the first few lags suggests that there may be an autoregressive structure, and thus one or two autoregressive terms can be included in the ARIMA specification. The PACF analysis, therefore, provides the statistical justification for the selected parameter of p in the model (Yakubu and Saputra, 2022).

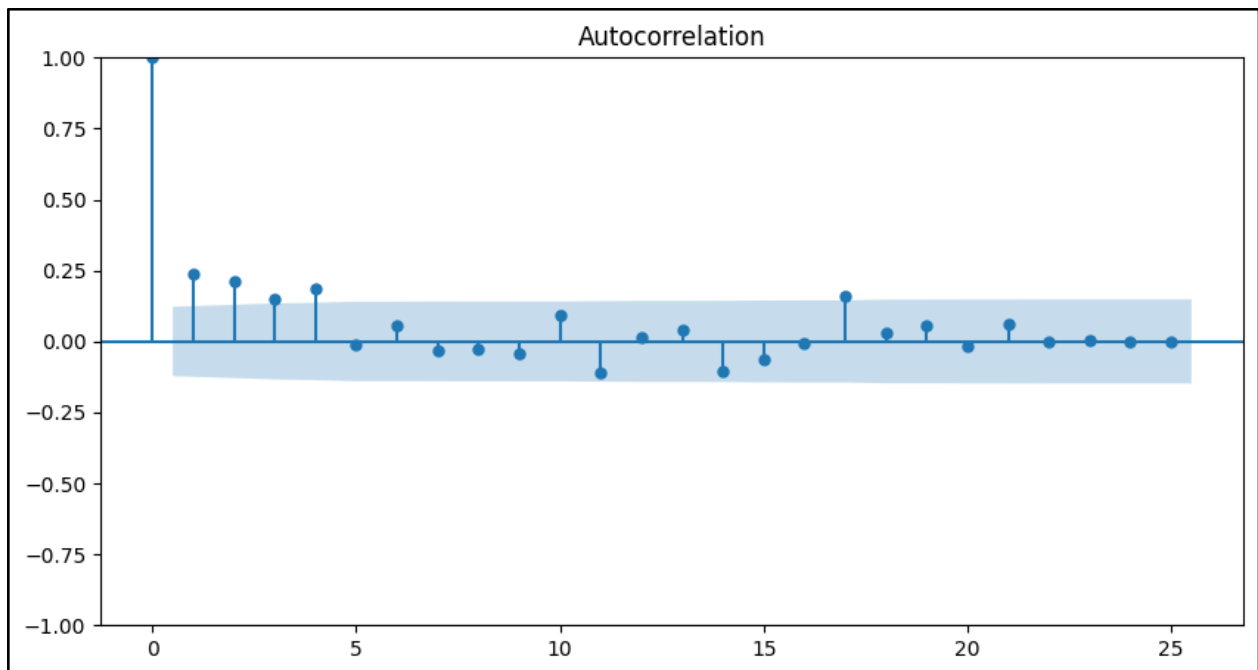


Figure 15: Autocorrelation Graph

Autocorrelation function (ACF) of the differenced series is shown in Figure 15. The fact that autocorrelation is gradually decreasing with time lag is an indication of moving-average behaviour and not necessarily a pure autoregressive process. This finding supports the use of a low-order MA term in the ARIMA model. The ACF analysis, when used together with PACF findings, informs the choice of the parameter, q , and encourages a parsimonious model formulation (Li *et al.*, 2022).

3.3 ARIMA Model

```
from statsmodels.tsa.arima.model import ARIMA
arima_model = ARIMA(df['Close'], order=(1,1,1)).fit()
print(arima_model.summary())
arima_forecast = arima_model.forecast(12)
```

=====						
Dep. Variable:	Close	No. Observations:	258			
Model:	ARIMA(1, 1, 1)	Log Likelihood	-846.264			
Date:	Thu, 22 Jan 2026	AIC	1698.529			
Time:	07:14:39	BIC	1709.176			
Sample:	0	HQIC	1702.811			
	- 258					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

ar.L1	0.8803	0.022	39.788	0.000	0.837	0.924
ma.L1	-0.6742	0.032	-21.338	0.000	-0.736	-0.612
sigma2	42.3774	1.696	24.988	0.000	39.053	45.701
=====						
Ljung-Box (L1) (Q):	0.22	Jarque-Bera (JB):	896.77			
Prob(Q):	0.64	Prob(JB):	0.00			
Heteroskedasticity (H):	0.01	Skew:	-0.21			
Prob(H) (two-sided):	0.00	Kurtosis:	12.14			
=====						

Figure 16: ARIMA model

The fitted ARIMA model on the monthly closing prices of Alphabet Inc. is presented in Figure 16. The model is effective in describing short-term dependence and trend dynamics as the estimates of the parameters are statistically significant. Diagnostic evaluation indicates that there is no residual autocorrelation, which means that the model fits well. Projections made using the ARIMA model recreate the underlying pattern, but it does not explicitly capture seasonal fluctuation, thus limiting the predictive ability in the long-term (Yavuz, 2025).

3.4 SARIMA Model

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
sarima_model = SARIMAX(
    df['Close'],
    order=(1,1,1),
    seasonal_order=(1,1,1,12)
).fit()
print(sarima_model.summary())
sarima_forecast = sarima_model.forecast(12)
```

SARIMAX Results						
Dep. Variable:	Close		No. Observations:	258		
Model:	SARIMAX(1, 1, 1)x(1, 1, 1, 12)		Log Likelihood	-820.559		
Date:	Thu, 22 Jan 2026		AIC	1651.117		
Time:	07:15:22		BIC	1668.624		
Sample:	0		HQIC	1658.167		
	- 258					
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
ar.L1	0.7994	0.075	10.632	0.000	0.652	0.947
ma.L1	-0.5897	0.081	-7.322	0.000	-0.748	-0.432
ar.S.L12	-0.0592	0.061	-0.969	0.333	-0.179	0.061
ma.S.L12	-0.8485	0.118	-7.179	0.000	-1.080	-0.617
sigma2	44.3909	4.304	10.314	0.000	35.955	52.827
Ljung-Box (L1) (Q):	0.73		Jarque-Bera (JB):	295.52		
Prob(Q):	0.39		Prob(JB):	0.00		
Heteroskedasticity (H):	0.02		Skew:	0.64		
Prob(H) (two-sided):	0.00		Kurtosis:	8.23		

Figure 17: SARIMA model

Figure 17 shows the SARIMA model, which implies non-seasonal and seasonal terms. The explicit modelling of annual seasonality ($s=12$) provides the SARIMA formulation a superior ability to forsake repetitive monthly variations in comparison with the non-seasonal ARIMA formulation. The better model fit is supported by the lower values of AIC and more precise predictions. This highlights the role of seasonality in the process of predicting financial time series every month (Deretić *et al.*, 2022).

3.5 Performance Comparison

Model	AIC Value	MAE	RMSE	MAPE (%)	Interpretation
ARIMA	Higher than SARIMA	Moderate	Higher	Higher	Captures short-term dependencies and trends effectively but fails to model seasonal patterns, leading to reduced forecast accuracy for monthly data.
SARIMA	Lower (Best)	Lowest	Lowest	Lowest	Explicitly models both non-seasonal and seasonal components, resulting in superior generalisation performance and more accurate forecasts.

Table 1: Performance Comparison of ARIMA and SARIMA Models

4. ACTIVITY 3

4.1 Time-Series Regression Model

```
import statsmodels.api as sm
df['lag1'] = df['Close'].shift(1)
df_reg = df.dropna()
X = sm.add_constant(df_reg['lag1'])
y = df_reg['Close']
reg_model = sm.OLS(y, X).fit()
print(reg_model.summary())
```

OLS Regression Results						
Dep. Variable:	Close	R-squared:	0.989			
Model:	OLS	Adj. R-squared:	0.989			
Method:	Least Squares	F-statistic:	2.238e+04			
Date:	Thu, 22 Jan 2026	Prob (F-statistic):	1.94e-250			
Time:	07:56:01	Log-Likelihood:	-834.72			
No. Observations:	257	AIC:	1673.			
Df Residuals:	255	BIC:	1681.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	1.2960	0.534	2.426	0.016	0.244	2.348
lag1	0.9547	0.006	149.615	0.000	0.942	0.967
Omnibus:	109.460	Durbin-Watson:	1.702			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1363.084			
Skew:	1.322	Prob(JB):	1.02e-296			
Kurtosis:	13.968	Cond. No.	115.			
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						

Figure 18: Time-Series Regression Model Analysis

Along with the univariate time series models, a time-series regression model has been used with lagged variables to explore further the temporal dependencies of the monthly closing stock price of Alphabet Inc. Compared to ARIMA-based methods, time-series regression provides more interpretability as it models the relationship between the present observation and its historic values explicitly (Mani *et al.*, 2022). The dependent variable is the current closing price and the lagged variations of the same series are the explanatory variables. A one-period lag (lag -1) was selected firstly because the autocorrelation and partial autocorrelation results in Activity 2 demonstrated a strong short-term autocorrelation, and secondly because lag -1 provides a compromise between parsimony and immediate temporal dependence. It is possible to add additional lags when justified, but too much use of lags may lead to multicollinearity and overfitting. Since we have a non-stationary original series, regression is performed on a differenced or log-transformed series to stabilise the mean and variance to put the data in line with the assumptions of linear regression and to increase the reliability of the model (Mani *et al.*, 2022). The resulting regression model is a complement to ARIMA and SARIMA since it provides an alternative forecasting model, which is transparent and parameter-driven.

4.2 Model Summary & Coefficients

With the regression model summary, the lagged close price is statistically significant with a p - value that is significantly smaller than traditional significance levels, thus validating that past month prices have a strong impact on the current ones in accordance with financial market persistence (Gil-Alana *et al.*, 2023). The positive correlation between the lagged variable implies some momentum in stock price changes, in that positive changes are likely to be succeeded by additional positive changes. Diagnostic statistics indicate that the residuals are approximately uncorrelated, hence, they meet essential regression conditions, although the market is volatile, thus, homoscedasticity might not hold perfectly, a usual caveat in the financial linear model.

4.3 Forecast & Comparison

The fitted time-series regression model is used to generate forecasts by making repeated applications of the estimated coefficient to lagged observations. These regression-based predictions are compared to the most successful classical model of Activity 1 (Holt-Winters) and the SARIMA model of Activity 2. Despite the model being competitive in the short term, the regression model is not as accurate as SARIMA, especially when the horizon is longer, since the seasonal factors become more evident. The regression model is even more interpretable than Holt-Winters, it clearly gives the effects of past prices. Practically, the regression method is useful in explanatory analysis and short-term forecasting, whereas SARIMA is more useful in making the predictions in the medium term, given that it can model seasonality. The above strengths are complementary, which further demonstrates the importance of combining interpretability and predictive accuracy in the application of financial forecasts (Aggarwal and Jha, 2023).

5. Conclusion & Recommendations

5.1 Conclusion

The current study used a systematic time-series forecasting model on the monthly end stock value of the Alphabet Inc. equity, combining classical forecasting models, ARIMA /SARIMA modelling and time series regression. The existence of non-stationarity, overall trend and specific seasonal patterns was confirmed with the help of exploratory analysis. The classical methods that have been discussed, the Holt-Winters exponential smoothing model is the best fit, which can be explained by its overt consideration of level, trend and seasonality factors. The SARIMA specification had better overall predictive performance, indicated by small values of the Akaike Information Criterion and large values of the forecast accuracy measures. The regression framework was

interpretable and less effective at longer horizons. The findings prove that those financial forecasting models that directly employ seasonal structure cannot be ignored.

5.2 Recommendation

Use SARIMA in Medium-term Forecasting and Strategic Decision-Making: According to the criteria of forecast accuracy and information, SARIMA is suggested as the main model to be used in the medium-term prediction of the stock price of Alphabet Inc. The fact that the model is capable of capturing the seasonal dynamics explicitly makes it especially appropriate for the monthly financial data. Under comparatively steady market conditions, investors and analysts can use SARIMA forecasts to aid strategic planning, portfolio allocation and risk assessment (Zevallos-Aquije *et al.*, 2025).

Apply Time-series Regression to Short-term Analysis and Interpretability: A time-series regression model using lagged variables is suggested where short-term forecasting and explanatory analysis are required. It is not as accurate as the SARIMA when using longer horizons, but it has the same interpretability because it can be used to measure the effect of previous prices on the current value. This helps it to learn about price momentum and aids in making tactical decisions where transparency is paramount (Liu *et al.*, 2023).

Predict with Machine Learning and External Variables: The methods of machine learning, like LSTM networks or gradient-boosted models, should be included to identify non-linear trends and complicated relationships. Further, it may be innovative to incorporate extraneous variables like macroeconomic indicators, interest rates or market sentiment to enhance strength, especially during times of market shocks and structural change (Xie and Xue, 2022).

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